

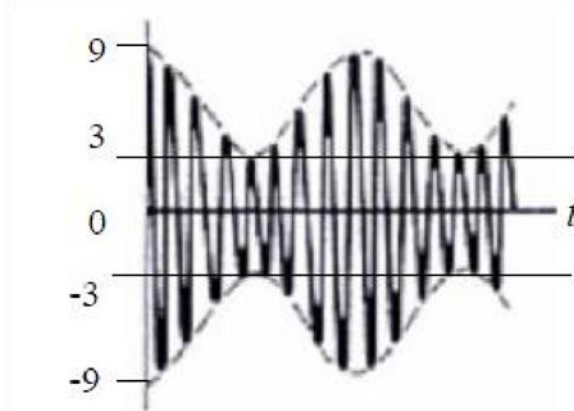


CNS2105 DATA AND VOICE COMMUNICATIONS CAT 2

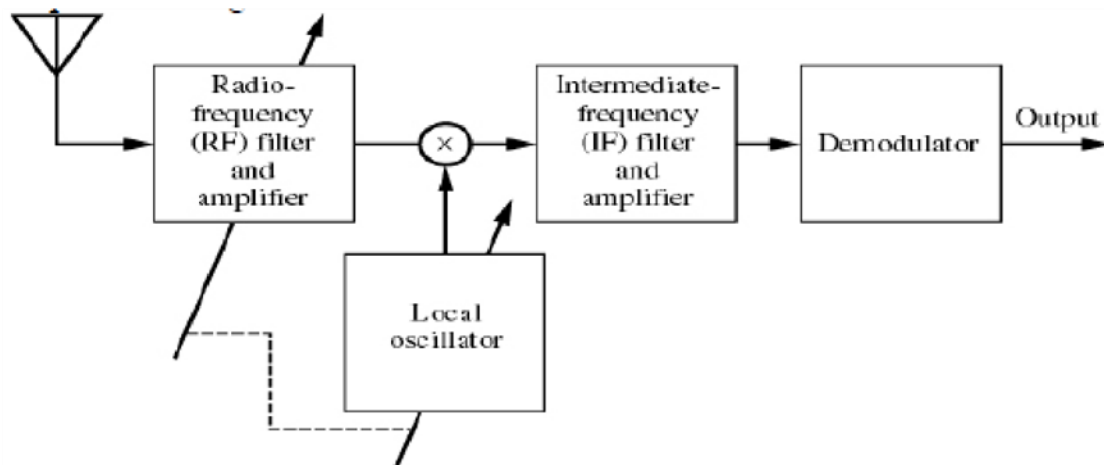
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10.15-11.15 HRS

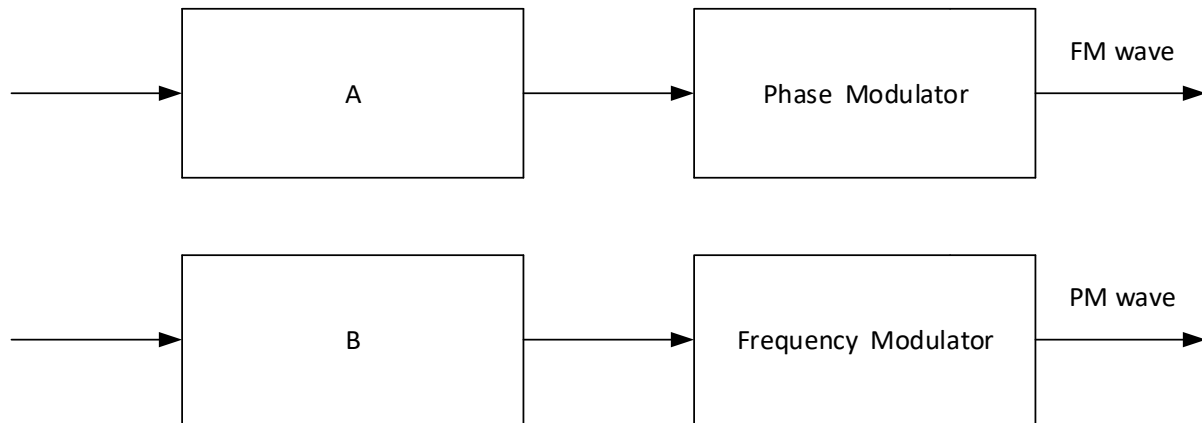
- 1) A sinusoidally modulated ordinary AM waveform is shown below



- a) Determine the modulation index.
 - b) Calculate the transmission efficiency.
 - c) Determine the amplitude of the carrier, which must be added to attain a modulation index of 0.3.
- 2) The output signal from an AM modulator is: $s(t) = 5\cos(1800\pi t) + 20\cos(2000\pi t) + 5\cos(2200\pi t)$
- a) Determine the modulation index.
 - a) Determine the ratio of the power in the sidebands to the power in the carrier.
- 3) A radio receiver used in the AM system is shown below. The mixer translates the carrier frequency f_c to a fixed IF of 455kHz by using a local oscillator of frequency f_{LO} . The broadcast-band frequencies range from 540 kHz to 1600kHz. Determine the range of tuning that must be provided in the local oscillator
- a) when f_{LO} is higher than f_c (superheterodyne receiver) and
 - b) when f_{LO} is lower than f_c .



- 4) A voice signal in the frequency range of $0 - 4\text{kHz}$ is sampled at the Nyquist rate. It is transmitted through a PCM system a signal to quantization noise ratio of 40 dB is desired. Determine each of the following:
 - a) Number of bits required to encode each bit
 - b) Number of quantization levels
 - c) Bit rate
 - d) Bandwidth occupied by the PCM signal
- 5) (a) Distinguish between frequency modulation and phase modulation techniques
 (b) A 6V 3 MHz sinusoidal carrier is frequency modulated by a 2.5 400Hz audio sine wave and the resulting deviation is 7.5 KHz . Determine the
 - i. Modulation index
 - ii. Bandwidth of the modulated signal
 - iii. Sensitivity of the modulator
- 6) A sinusoidal signal of amplitude 100V is amplitude modulated by a carrier signal to a depth of 25% . Determine the
 - a) Maximum and minimum voltage
 - b) Modulating signal voltage
 - c) Amplitude of the lower side frequency
- 7) A 20V , 150MHz carrier wave is frequency modulated by a 2000 Hz tone when the frequency deviation is 1050Hz . Determine the
 - a) Modulation index
 - b) Carrier amplitude
- 8) The figures below show alternative methods of generating PM and M waves. Explain the operations performed by blocks A and B



- 9) You are given a modulated waveform $s(t) = A\cos(2\pi f_a t + a\sin 2\pi f_b t)$. If this is a PM wave, write down the:
- a) Message signal $m(t)$
 - b) Phase deviation
 - c) Unmodulated frequency of the carrier
 - d) Frequency of the message signal $m(t)$